DBMS Project Guidelines

The project's goal is to develop a database management system tailored to a specific application. You are suppose to work in a team of 3 students from the same lab group. Your initial task is to select a real-time database application in line with current trends. Avoid opting for common choices like library, railway, airplane, university management systems, etc.

The following points has to be covered:

1. Project Statement: give the brief overview of project, use cases (applications) and objective.
2. Entity Relationship Diagram (ERD): Recognize the entities, attributes, keys, relationships, cardinalities, participation constraints, and generalization/specialization. Utilize a tool to create the ERD; hand-drawn diagrams on paper will not be considered.
3. Transform your ER model into a relational model and illustrate a schema diagram accordingly. Specify the tables along with their attributes and constraints, including domain, primary key, referential integrity, not null, unique, check, etc.
4. Normalization: Determine the functional dependencies present in the tables and utilize this information to normalize your relational model up to 3NF. Clearly demonstrate all the steps involved, including the final set of decomposed relations.
5. Implement your normalized relational model in SQL.
    a. Create all the tables along with all necessary integrity constraints.
    b. Populate tables with some relevant data values. (You may create forms i.e., front end for inserting the data into tables atleast in hundreds).
    c. Write various queries for the retrieval of data related to the application. (You may also use forms as a GUI for submitting the queries and displaying the results.

        i. DDL Queries- Create (with integrity constraints), Drop, Alter, Truncate, and Rename
        ii. DML Queries- Insert, Update (with and without cascade), and Delete (with and without cascade)
        iii. DCL Queries- Grant and Revoke
        iv. TCL Queries- Commit, Savepoint, and Rollback
        v. DQL Queries- Select
            1. with where
            2. aggregate
            3. in, like, distinct
            4. orderby, limit, and offset
            5. groupby, groupby with having
            6. joins all types
            7. union, union all, intersect, and except
            8. except, any, and all
        vi. Subqueries
        vii. Views and Materialized Views
        viii. Stored Procedures, Functions, and Triggers

NOTE: The front-end part is optional as well as bonus. If you are using this part then you can use C/C++/Java/Python as high-level language for embedding the SQL queries and creation of forms. You also need to understand the connectivity of back end with front end.